

Balancing Random Pokémon Battles

Executive Summary

T. Daniels, X. Huang, G. Knapp, M. Mannan, M. Newman

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INTRODUCTION

If you had to build two teams of 6 Pokémon to battle, how do you give both teams a fair chance? With over 1,000 Pokémon to choose from, each with different types, abilities, move-sets, and stats, the number of possible teams is astronomical. What if you had to do this *randomly*? This is exactly the issue that the developers of [Pokémon Showdown](#) (“Showdown”) grapple with in curating their most popular game format: the *Gen-9 Random Battle*.

Ideally, random teams should be balanced so that both players have even chances of winning at match start. Our work in this project tries to analyze just how “well-balanced” (at least, roughly!) these random teams are, by seeing if we can predict a battle’s winner just using the matchup data at battle start.

DATA COLLECTION

Showdown stores text logs of all the battles played on their site. However, these text logs can be difficult to parse, and can significantly vary in content, so to collect and handle them we developed our custom “Battle” class. In total, we saved and parsed about 18,000 Showdown match replays.

The Team Computation Hurdle

In random battles, players’ individual Pokémon are not logged unless they are fielded. In order to learn the complete team rosters in each battle—including those Pokémon not fielded—it was necessary to 1) locate the pseudorandom seed-strings used in generating players’ random teams; and 2) feed these into a custom interface with Showdown’s source code to re-generate and record the complete match rosters.

FEATURE ENGINEERING

There are a large number of features to consider in modeling teams of six Pokémon. In addition to Pokémon's *base stats* (like HP or Attack) and players' *Elo ratings* (see either [Ratings](#) or [Wikipedia](#)), and other team-based statistics, we introduced the **Advantage** statistic for 1v1 Pokémon encounters and the aggregate **Team Advantage** statistic for our model training. The Advantage statistic is detailed in [AdvantageStat](#).

While Team Advantage was our most impactful feature, it has known drawbacks, such as failing to consider the specific moves or abilities for each Pokémon. In an attempt to address some of these drawbacks, we also considered the **Stat-Booster Differential**—the difference in the numbers of Pokémon on each team having "stat-boosting" moves or abilities.

Moreover, newer versions of the Advantage stat that incorporate more of Showdown's battle source code have been developed (and are continuing to be developed).

MODEL SELECTION

For a "baseline" model we used a single-feature predictor based on players' Elo rating differences, since the Elo system maps this difference to estimated probabilities of each player winning. Beyond this baseline, we considered Logistic Regression models both with and without intercept terms (the "biased" and "unbiased" models, respectively), the standard Decision Tree and Random Forest models, and the boosting models HGBost and XGBost.

We trained these models on the features which optimized their cross-validation accuracy: for the unbiased Logistic Regression, those features were Elo differential, Total Advantage, and Stat-Booster Differential. The remaining models trained on all of those features plus type diversity differential and total stat differential.

Accuracy was chosen over other metrics for its ease of interpretation and because our application does not require special precautions against "false positives and negatives", such as applications in, e.g., medical diagnosis. Additionally, we wanted to select a model that was well-calibrated: since Player 1 lost in 52% of the matches, we wanted a model which would predict that Player 1 would lose approximately 52% of the time.

On our training data, the baseline- and unbiased Logistic Regression models were 52.1% and 53.1% accurate, and both were well-calibrated. The remaining models had accuracies all close to the former two, but were poorly calibrated, leading us to select the unbiased logistic regression as our final model.

CONCLUSION AND NEXT STEPS

Conclusion

After selecting the unbiased Logistic Regression as our final model and trialing it on our test data, our model achieved an accuracy of 53.6%. As this is not much better than a coin flip's accuracy of 50%, we conclude that the level and move-set scaling performed by Pokémon Showdown for Gen-9 random battles is sufficiently balanced to maintain interesting and unpredictable gameplay.

Next Steps

There remains potential to create a predictive model based on turn-by-turn analysis. Instead of relying on statistics determined pre-game, this model could estimate player win probabilities as a function of the game “state”, updating its predictions after every turn. In addition, the Advantage metric can be further refined to better account for Pokémon's special abilities and held items.